

Customer Churn Data Analytics Project

Stage 4 Summary: Data Quality Assessment

This document summarizes the Data Quality Assessment stage of the Customer Churn Data Analytics Project, based on the diagnostic notebook *04. Data Quality Assessment*. The purpose of this stage is to systematically assess whether the raw dataset is complete, valid, consistent, unique, and structurally reliable — before any cleaning is performed.

Distinction from adjacent stages

- Stage 3 — Data Understanding: What does the data look like?
- Stage 4 — Data Quality Assessment: What is wrong, missing, inconsistent, or questionable about the data?
- Stage 5 — Data Cleaning & Preparation: What actions should we take to fix those issues?

This notebook diagnoses problems only. The raw dataset is not modified at this stage; all numeric conversions and derived masks used for checks (e.g., coercing `TotalCharges` to numeric) are temporary and used strictly for diagnosis.

1. Stage Activities

Step	Dimension	Description
4.1	Completeness	Identify missing values/nulls across all columns.
4.2	Blank / Empty Values	Detect empty strings or whitespace-only values not represented as standard nulls.
4.3	Data Type Validity	Check whether each variable is stored using an appropriate data type.
4.4	Domain Validity	Check whether categorical variables contain only expected/valid values.
4.5	Numerical Validity	Check ranges and identify impossible or suspicious numerical values.
4.6	Uniqueness	Check whether <code>customerID</code> is unique and identify duplicate records/customer IDs.
4.7	Consistency	Identify inconsistent representations, naming, capitalization, spacing, or category encoding.
4.8	Cross-Field Integrity	Check logical relationships between related variables.
4.9	Outlier Assessment	Identify potentially unusual numerical observations and whether they require investigation.
4.10	Target Integrity	Check whether <code>Churn</code> contains valid and expected values and whether any target values are missing.
4.11	Data Quality Summary	Summarize every detected quality issue, its severity, affected columns/records, and recommended treatment.

2. Diagnostic Workflow (Notebook Overview)

The accompanying notebook loads the raw CSV file into a working DataFrame (df) and runs a series of independent audits, each targeting one dimension of data quality:

Completeness & Blank-Value Audits

- Missing-value audit: counts and percentages of standard Pandas nulls (NaN) across all columns.
- TotalCharges investigated separately: its dtype and tail values are inspected, since it may contain blank strings rather than true NaN values — a case standard null checks alone would miss.
- Non-numeric TotalCharges values are detected by coercing the column to numeric with errors="coerce" and counting resulting NaNs; affected records are cross-referenced against tenure, MonthlyCharges, Contract, and Churn to understand business context before any fix is decided.

Uniqueness Audits

- Exact duplicate rows are counted (not removed).
- customerID uniqueness is checked: missing IDs, unique ID count vs. total row count, and duplicated ID count are assessed separately.
- Any duplicate customerID records found are inspected in full before any interpretation or removal decision is made.

Consistency & Domain Validity Audits

- Unique values for all major categorical fields (gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, Contract, PaperlessBilling, PaymentMethod, Churn) are reviewed for unexpected labels or spelling variants.
- A whitespace audit checks each categorical field for leading/trailing whitespace in values, counting (not correcting) affected entries.
- A cardinality audit compares the number of unique values per categorical field to flag columns whose category count looks larger than expected.

Target Integrity

The Churn column is validated for expected values and checked for missing entries, since nearly every downstream business question depends on the correctness of this field.

Numerical Validity & Outlier Assessment

- Negative-value checks are run on tenure, MonthlyCharges, and the temporarily coerced numeric TotalCharges, to catch logically impossible values.
- An IQR-based outlier check is applied to MonthlyCharges (bounds = $Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$) to flag potential outliers as candidates for review — not automatically invalid records.

Cross-Field Integrity (Business-Rule Checks)

Logical relationships between related service fields are validated semantically, not just checked for missing values:

- Phone-service consistency: customers with PhoneService = "No" are expected to have MultipleLines = "No phone service"; mismatches are counted.
- Internet-service consistency: customers with InternetService = "No" are expected to show "No internet service" across OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies; mismatches are counted.

Compact Quality Snapshot

A final consolidated snapshot brings all checks together in one place: row/column counts, total missing cells, exact duplicate rows, duplicate customer IDs, invalid TotalCharges count, negative-value counts for tenure/MonthlyCharges/TotalCharges, and the two cross-field inconsistency counts.

3. Data Quality Issue Log (to be completed with notebook output)

The table below provides the structure for consolidating findings once the notebook is run against the actual dataset. Each row should be completed with the observed count/percentage, an assessed severity, and a recommended treatment for Stage 5 — Data Cleaning & Preparation.

Issue	Affected Column(s)	Count / %	Severity / Treatment
Missing values (NaN)	All columns	—	TBD
Non-numeric TotalCharges	TotalCharges	—	TBD
Exact duplicate rows	All columns	—	TBD
Duplicate customer IDs	customerID	—	TBD
Whitespace in categorical values	Categorical fields	—	TBD
Unexpected categories / cardinality	Categorical fields	—	TBD
Missing/invalid Churn values	Churn	—	TBD
Negative tenure / charges	tenure, MonthlyCharges, TotalCharges	—	TBD
MonthlyCharges outliers (IQR)	MonthlyCharges	—	TBD
Phone-service inconsistencies	PhoneService, MultipleLines	—	TBD
Internet-service inconsistencies	InternetService + dependent service fields	—	TBD

This log should be populated with the actual output values from running the notebook against the dataset, since specific counts and percentages are not fabricated in this summary document.

4. Deliverable

A consolidated Data Quality Summary — covering every detected issue, its severity, the affected columns/records, and a recommended treatment — to be documented in Excel/Word and used directly as the input to Stage 5 — Data Cleaning & Preparation.